

# Wantian Huang

*Economist: empirical IO, demand estimation and pricing, causal inference and experimentation*

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## EDUCATION

Ph.D. Economics, University of Maryland, College Park

Expected May 2027

M.A. Economics, Columbia University

2020

B.S. Economics & Mathematics, University of California San Diego

2018

## PROFESSIONAL EXPERIENCE

### Economist Intern, Amazon Ads Marketing Finance

Seattle, WA Jun 2026 – Aug 2026

- Evaluated five randomized promotions in an advertising-credit program and showed that credits raised seller ad adoption without incremental ad revenue, with returns varying widely across segments; estimated intent-to-treat effects by covariate-adjusted OLS, with difference-in-differences and event studies for dynamics under repeated offers.
- Found that effects fade with repeated offers and concentrate among sellers with high predicted take-up (causal forests, compliance-score matching, double machine learning), supporting capped and targeted offers.
- Authored the technical review and briefed the Ads finance director; quantified millions of dollars in annual credits with no incremental return and recommended credit-menu and re-offer changes plus a follow-on experiment.

### Research Assistant to Prof. Chenyu Yang, University of Maryland

Summers 2024 – 2025

- Built and analyzed large datasets (Nielsen RMS/HMS, streaming analytics) in Python, Stata, and MATLAB; implemented econometric and structural models of demand estimation, merger simulation, and counterfactual analysis.

## RESEARCH

### Testing the Substitution Predictions of Demand Models: Evidence from the Blue Bell Recall

*Job Market Paper*

- Use the 2015 Blue Bell recall, a whole-brand withdrawal, to test whether demand models estimated on pre-recall data predict how much of a removed brand's demand rivals retain and which rivals receive it; logit, nested logit, BLP, and RCNL with household micro moments on 3.7M Nielsen product-market-months.
- Find that rivals recaptured about three quarters of Blue Bell's volume, half of it to private label; models without nesting predict too little retention, while nests that pool private label with Blue Bell's mainstream tubs come close on both margins, so nest placement matters more than taste heterogeneity for merger and welfare predictions.

### Economies of Scope in Retail Distribution: A Study of Ready-to-Consume Categories

*with Ying Fan, Yidan Xu, and Chenyu Yang*

- Show that economies of scope raise product variety and consumer welfare, and that ignoring them understates the gains from small-firm mergers.
- Estimate demand with a BLP random-coefficients model and micro-moments on Nielsen RMS scanner data, recovering elasticities, diversion ratios, and marginal costs for merger simulation.
- Bound fixed costs under minimal assumptions using moment inequalities with optimal-transport inference; run estimation and counterfactual pipelines on an HPC cluster.

### Evaluating the Welfare Impacts of EV Charging Station Subsidies: The NEVI Program in Texas

- Assess cost-effectiveness of 50 subsidized stations: \$31.8M in fixed costs against \$15.2M in annual expected welfare gains, implying roughly a two-year payback.
- Estimate EV demand with structural models that accommodate prevalent zero market shares.

### Quality Uncertainty, Learning, and Product Variety in Live Streaming Platforms

*In progress*

- Quantify the welfare contribution of marginal streamers and evaluate counterfactual monetization, retention, and merger scenarios.
- Develop a nested two-stage demand framework for viewer engagement; model streamer learning, effort, and exit.

**Presentation:** Industrial Organization Workshop, University of Maryland (scheduled, October 2026)

## TEACHING (SELECTED)

Instructor: Economics of Regulation and Antitrust (Summer 2024)

Teaching Assistant, M.S. program (2022 – 2026): Microeconomic Analysis, Game Theory, Financial Economics

## TECHNICAL SKILLS

- Programming: Python (pandas, NumPy, scikit-learn, pyfixest, EconML), SQL (Redshift, Hive, Spark), Stata, MATLAB, LaTeX; Parquet panels, HPC (SLURM) and distributed computing
- Methods: A/B testing and RCTs, DiD (canonical and staggered), event studies, IV/RD, causal forests, double machine learning; BLP and nested-logit demand, merger simulation, moment inequalities, partial identification